All Abstract Fictional Worlds Exist, but Not All Are Sentient: Consciousness, Dreams and the Mathematical Universe

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Abstract

All abstract fictional worlds exist, since every consistent structure exists and some consistent math fixes each plot hole, but not all are sentient. A fictional world can be a real experience, because an experience and what it represents are different things: a dream is experienced even when its room, its people and its events exist nowhere outside it. Matheism identifies a mind with its mathematical organization, so the experience needs that organization at work and nothing from the represented world around it. A stronger claim follows from Matheism's realization premise that every consistent mathematical structure exists: a mind can exist with no host universe at all, its whole world being the one it represents. Not every story qualifies, since many are inconsistent or leave the mind's own organization unspecified; what counts is a consistent structure that experiences.

Keywords: mathematical monism, consciousness, fiction, representation, computational organization, mathematical universe hypothesis, observer measure

1. Introduction: The Fiction Question

An experience of a fictional world is real, because the experience is the mind's own organization at work and the world it represents is only its content. A dreamed room need not exist outside the dream for the experience to occur. A story, a world matching the story, and a mind imagining it are three different

things. This paper asks what Matheism implies about the third: an experiencing structure whose represented surroundings need not be its actual surroundings.

Matheism argues that whatever affects an experiencing system must enter determinate relations with it, and then makes an identity claim: the complete structure of those relations is the system. Fictions commonly leave their laws and details unspecified. An unspecified detail is no proof of inconsistency, since a sketch may admit many consistent completions.

A fiction can be real as experienced content while no external setting matches it. Anyone extending that observation to every proposed fiction has to show that a suitable experiencing structure exists in each case. The argument below therefore draws that implication conditionally, and it does not read an unlimited repertoire of experience out of the word *imaginable*.

The question sits in a larger chain:

1. Everything that exists is ruled or random, and both are mathematical, so physics is mathematics, and anything outside mathematics would be outside logic. That is an a priori proof.
2. Every consistent structure exists as mathematics, fictions included, and every plot hole is fixed by some consistent math. What exists physically is what logical steps in mathematics produce. Mathematics cannot fail to exist, so there is something rather than nothing.
3. Each bit of a description that no rule decides is a 50/50, so each added bit halves the description's weight. Short laws dominate, and a universe like ours starts simple, in its laws and its state, as observation also shows: the known laws fit on a page, and the early universe was nearly smooth.
4. Observers like us are complex, so a simple start has to grow complexity, as simple rules do.

5. A cause produces its effect as the next state, and a continuum has no next state, so the growth comes in discrete causal steps. That is an a priori proof in the logician's sense.

6. Time is those steps, and space is how far they spread, at a base rate, the speed of light. An endless chain of steps has no last step to run back from, so time runs forward. That too is a priori. A history run backwards would also have to state the step where it turns back, and each bit of that timestamp halves its weight.

7. Observers arise in forward time, where complexity grows, and their minds run forward with it: a mind models its world from records of steps already taken, and each thought is produced from the one before, so no one remembers backwards from the end.

8. A mind is that growth at work, and its experience lives in its transitions, so the hard problem of consciousness, which assumes that feeling is something other than structure, does not block the question. The bare form of being a subject, with memories and traits set aside, is the same in every mind: one self.

Each link stands on its fact, so a reason can fail without taking the chain with it: if the weight on short descriptions were not why our laws are simple, they would still be simple.

2. The Transitions Thesis

A process can be mathematically consistent and still contain mistaken memories and conflicting beliefs. Being right about the world is no condition of being a consistent process. A candidate computational account therefore needs defined states and transition rules, and it has to say how representations, memory and access are implemented.

That distinction allows a mind to represent violations of its surrounding laws. A mind can carry a label for an impossible object, and carrying the label does not make the mind realize every property the label assigns. Identifying experience with computational organization is a substantive monist proposal, one treating mind and matter as one kind of thing. A successful version must distinguish the organizations that experience from those that merely contain descriptions of experience. The hard problem of consciousness, the question of why physical processes give rise to experience, does not block the proposal: it assumes feeling is something other than structure, and any account of what feeling adds would itself be structure.

3. What Dreams Establish

A dream of flying can occur while the dreamer remains in bed. The represented flight and the process representing it have different physical descriptions. The dream process can be internally well defined even when its content conflicts with waking events. This separates consistency of the process from truth of its content.

The dream shows that an experience can be real with no matching event outside it. It does not show that the dream process runs free of all supporting physical organization: the sleeper still has a brain. Mathematical monism, the view that reality is wholly mathematical structure, then faces a further question. Can the relevant organization be specified as a complete structure with no larger biological setting around it?

4. Standalone Minds and the Realization Premise

A mind identified with its organization can be implemented in more than one way without acquiring a further kind of substance. Suppose a complete computational

structure has exactly the organization sufficient for experience. A different substrate, the material a mind runs on, would not need an additional substance if it preserved everything relevant. Which relations must be preserved is part of the proposal, not something settled by matching a short output trace.

Matheism adds a realization premise: admissible mathematical structures exist without an external selector. Standalone means that the experiencing organization needs no separate world corresponding to its represented setting. Its complete specification must still include whatever states, relations, transitions and boundary information (its starting and edge conditions) make it the proposed mind.

The central inference follows: the realization premise makes the specified structure exist, and a structure with the organization sufficient for experience has that experience with no additional nonstructural bearer. The represented world can remain fictional.

5. Shared Dreams

Coupling two systems does not fuse them into one subject. Two systems can exchange signals through shared variables and transition rules, and that gives a coupled system, nothing more. A common simulated environment could supply coordinated inputs to two experiencing systems. Sharing a reference takes actual relations in the implementation, not merely two matching private descriptions.

This possibility does not imply that every external world consists only of minds. Mathematical monism permits structures without observers as well as structures with observers. The stronger claim that our universe is entirely constituted by connected experiences is an additional thesis, defended in different forms by Alfred North Whitehead's process philosophy (Whitehead, 1929) and Timothy

Sprigge's absolute idealism (Sprigge, 1983); it is not a consequence of the coupled simulation example.

6. The Fiction Question Reframed

A real experience of fictional content follows from three premises: a suitable structure exists, it has whatever organization experience takes, and one of its experiences represents the fiction. Stated precisely, the realization principle admits a structure S; S has the organization sufficient for experience; and an experience in S represents the content F. The conclusion concerns the experience of F. It does not make F true of the mind's surroundings.

Going from *a* fiction to *every* fiction adds an assertion: for every proposed F there exists a suitable S. No general construction for that is given here. A narrative that cannot be built as an external world might still be experienced as content. That second route has its own conditions, and it need not inherit every physical claim the narrative makes.

7. Fluctuation Observers: Formation, Experience and Weight

A Boltzmann observer, a mind assembled by a rare fluctuation of the kind the physicist Ludwig Boltzmann proposed, instead of by biological development (Boltzmann, 1895; Carroll, 2010), is not disqualified by its origin. If a fluctuation implements all the organization sufficient for experience, the organizational account attributes experience to it. Whether a proposed fluctuation does so remains a physical and philosophical question.

Whether branching keeps happening depends on the particular model, so the many-worlds dynamics of Hugh Everett (Everett, 1957), in which each possible quantum outcome occurs in its own branch, does not guarantee continual

branching in every state. The Hamiltonian (a system's energy function, which governs its evolution), the initial state and the development of correlated records matter. A description length comparison, in the tradition of Ray Solomonoff and Jürgen Schmidhuber (Solomonoff, 1964; Schmidhuber, 2000), has to give both candidates their shortest available generative account. A fluctuation need not be specified particle by particle if a cosmological generator produces it. The calculation must include how that event is selected and weighted.

A minimal duration for experience is not derived in this paper. The criterion has to be about the organization a system implements, not about how finely the notation slices it.

8. Dynamics, Gravity and Reachability

A mind we can write down in our mathematical language is not thereby reached by our wavefunction, the quantum state of our whole universe. A specified quantum cosmology supplies a state and dynamics. Branches share the fundamental dynamics; they may differ in effective parameters through different field states. A law can permit a state without the selected initial condition ever leading there.

The experience argument does not require a proof that Level III, the branches of our own wavefunction, contains Level IV, the full range of consistent mathematical structures, in Max Tegmark's four-level scheme (Tegmark, 1998, 2014). It concerns what a suitable experiencing structure has wherever it is realized. Whether our cosmology supplies that structure and with what measure, or probability weight, remains an independent task.

9. What Undecidability Establishes

To establish that every proposed fiction is experienced somewhere, one would need an account of which descriptions admit an experiencing implementation. Checking the descriptions one at a time is no route to it either, since for a rich enough description language no general procedure sorts the consistent descriptions from the rest. Stated precisely, for sufficiently expressive theories, no algorithm decides consistency in every case, a limit that goes back to Alan Turing's work on computable numbers (Turing, 1936). Neither point draws a boundary around possible consciousness by itself.

An undecidability theorem for recognizing experiencing structures would need an effective input language (one precise enough for a machine to process), a definite organizational criterion and a reduction from a known undecidable problem, a translation of that problem into this one. No such reduction is supplied here. Individual implementations may be specified and investigated even when no general decision method exists.

Timothy Williamson's discussion of knowing through imagination (Williamson, 2007, 2016) concerns how imagination can support modal knowledge, knowledge of what is possible or necessary. Kendall Walton treats works of fiction as props in games of make-believe (Walton, 1990). The computational proposal is about the activity that implements an actual experience. Fictional truth and the conventions governing interpretation are separate topics. On the monist account, the reality of an experience comes apart from the realization of the world it represents.

10. Connection to the Broader Program

The broader Matheist program does not establish the Standard Model (the accepted theory of the known elementary particles), cosmic matter asymmetry (the excess of matter over antimatter) or observer measures from a simplicity

prior (probabilities favoring simpler descriptions before any evidence) alone. Assess the standalone case by specifying its complete dynamics and the criterion it satisfies. Once the structure is specified, the realization premise makes it exist, and the monist account attributes experience without adding a host world in which the depicted story is true.

Imagination can represent mathematical possibilities as well as confused or incomplete descriptions. It is not evidence of a causal channel to another realized world. This argument is about the concrete process that does the experiencing. What that process represents is one thing; what would make the representation true outside it is another.

11. Conclusion

A represented world can be fictional while the experience representing it is real, because a mind is its mathematical organization and the world it represents is only content. A suitable mathematical mind supplies that experience, and Matheism's realization premise lets it exist with no host universe: admissible structures exist without an external selector. The same criterion for experience must apply whatever the mind's origin.

The represented room may be fictional. The experience of it is real.

References

Boltzmann, L. (1895). On certain questions of the theory of gases. *Nature*, 51, 413-415.

Carroll, S. (2010). *From Eternity to Here*. Dutton.

Everett, H. (1957). "Relative State" formulation of quantum mechanics. *Reviews of Modern Physics*, 29, 454.

Schmidhuber, J. (2000). Algorithmic theories of everything. *arXiv:quant-ph/0011122*.

Solomonoff, R. J. (1964). A formal theory of inductive inference. *Information and Control*, 7, 1-22, 224-254.

Sprigge, T. L. S. (1983). *The Vindication of Absolute Idealism*. Edinburgh University Press.

Tegmark, M. (1998). Is "the theory of everything" merely the ultimate ensemble theory? *Annals of Physics*, 270, 1-51.

Tegmark, M. (2014). *Our Mathematical Universe*. Knopf.

Turing, A. M. (1936). On computable numbers, with an application to the Entscheidungsproblem. *Proceedings of the London Mathematical Society*, 42, 230-265.

Walton, K. L. (1990). *Mimesis as Make-Believe: On the Foundations of the Representational Arts*. Harvard University Press.

Whitehead, A. N. (1929). *Process and Reality*. Macmillan.

Williamson, T. (2007). *The Philosophy of Philosophy*. Blackwell.

Williamson, T. (2016). Knowing by Imagining. In A. Kind and P. Kung (Eds.), Knowledge Through Imagination. Oxford University Press.